Definitions used in the simulations

Potentials. The attached head is held by a quadratic potential and the detached head moves in a harmonic well,

$$w_1(x) = \frac{k_1}{2}\,x^2, \qquad w_0(x) = \frac{k_0}{2}\,x^2, \qquad f = -\partial_x w_1(x) = -k_1\,x, \tag{1}$$

with $k_1 = 0.5$ pN/nm, $k_0 = 0.2$ pN/nm, damping $\eta = 0.2$ pN ms/nm, $k_B T = 4.1$ zJ and site spacing $d = 40$ nm (Fig. 1).

Rates. The rates enter the Poisson measures of (3.2); $h \in [-d/2, d/2)$ is the signed distance from the tip to the closest actin site ($h > 0$: the site is in front of the tip), recomputed at every time step from the window of size d centred on the tip. We take

$$K_{0\to1}(x,h) = k_{\rm att}\,\mathbf{1}_{\{h>h_*\}}, \qquad 0 \le h_* < d/2, \tag{2}$$

$$K_{1\to0}(x,h) = \frac{\lambda_{\rm det}}{d/2}\,\mathbf{1}_{\{-d/2<h<0\}} \tag{3}$$

$K_{0\to1}$ is a step in h: a detached head can bind only if the closest site is in front of the tip by more than h_* (here $h_* = 5$ nm). The limit $k_{\rm att} \to \infty$ makes the threshold alone decide: the head binds as soon as $h > h_*$. $K_{1\to0}(x,\cdot)$ is a density in the landing mark h: the total detachment rate is $\int K_{1\to0}(x,h)\,dh = \lambda_{\rm det}$ and the mark is uniform on $(-d/2, 0)$. At a $0 \to 1$ jump, $X \to X + h^\star$ and $h \to 0$; at a $1 \to 0$ jump, $X \to X - h^\star$ and $h \to h^\star$. Since an attached head moves with its site ($\dot X = \dot y$ and the sites move at $\dot y$), $dh = 0$ while attached: h cannot trigger a detachment, which is why $K_{1\to0}$ is a jump rate.

Macroscopic equation (soft device). With σ imposed,

$$\nu\,\dot y = N\sigma - \sum_{i=1}^{N} \alpha_t^i\,\partial_x w_1(X_t^i), \tag{4}$$

and $\nu_{\rm ph} = \nu/N$ ($= 0.05$ pN ms/nm in the simulations) is the viscosity per head.

Consequences, case $N = 1$.

- *Detached head*: $\tau_c = 0$, hence $\dot y = \sigma/\nu$ and h increases on average with slope σ/ν.
- *Attached head* (w_1 quadratic): $\dot X = \dot y = (\sigma - k_1 X)/\nu$, hence

$$X(t) = \frac{\sigma}{k_1} + \Big(X(t_a^+) - \frac{\sigma}{k_1}\Big)e^{-k_1(t-t_a)/\nu}, \qquad X(t_a^+) = X(t_a^-) + h, \tag{5}$$

  an exponential (not a linear) evolution toward σ/k_1, with rate k_1/ν.
- *No detachment* ($\lambda_{\rm det} = 0$): the head stays attached, $\dot y \to 0$, $\tau_c \to \sigma$, $X \to \sigma/k_1$: the system stops.
- *No attachment* ($k_{\rm att} = 0$): $\tau_c = 0$ and $\dot y = \sigma/\nu$ for all times, linear in σ.

Between two jumps, any N. Differentiating the macroscopic equation, with $\dot\sigma = 0$ and $\dot X^i = \dot y$ for the N_a attached heads,

$$\nu\,\ddot y = -k_1 N_a\,\dot y \quad\Longrightarrow\quad \dot y(t) = \dot y(t_j^+)\,e^{-k_1 N_a (t-t_j)/\nu}, \tag{6}$$

$$\dot y(t_j^+) = \frac{1}{\nu}\Big(N\sigma - \sum_i \alpha^i\,k_1 X^i_{t_j^+}\Big). \tag{7}$$

The decay rate depends only on the number of attached heads, and σ enters through the value of $\dot y$ right after the jump, which is fixed by the force balance. The speed stops ($\dot y \to 0$) when $\tau_c = \sigma$.

Figure 1: Potentials w_0, w_1, forces, attachment rate $K_{0\to1}(h)$ (step at h_*) and detachment density $K_{1\to0}(h)$.